Jeremy Hui

(848) 213-2934 | jeremywhui@gmail.com | https://jeremywhui.github.io/

Education

Rutgers University - New Brunswick **Sep. 2022 - May 2026**
Bachelor of Science in Electrical & Computer Engineering and Computer Science Double Major, Mathematics Minor
GPA: 4.0/4.0 (US conversion)

Research Experience

Hardware Secure Experience Lab (HWSEL) **Sep. 2024 - Present**
Supervisor: Prof. Sheng Wei *(Rutgers University)*
Automated Secure Deployment and Runtime Monitoring for Confidential Containers

- Implemented disaggregated memory prototypes through leveraging hardware-level isolation features on Arm Confidential Compute Architecture processors.
- Designed and built an automated deployment agent to deploy Confidential Containers (CoCo) for Deep Neural Networks workloads and identify malicious behaviors.
- Deployed CoCo on non-confidential hardware, and analyzed 10 microarchitectural performance counters to detect for indicators of compromise in 5 workloads.
- Developed extension to Kata Agent to help non-security researchers secure confidential workloads, which creates sandboxed environments from textual descriptions.

Princeton-Intel Research Experience for Undergraduates **Jun. 2024 - Jul. 2024**
Supervisor: Prof. David Wentzlaff *(Princeton University)*
Performing Rowhammer-Like Attacks on OpenSSD NAND Flash

- Identified and documented methods to bypass Bose–Chaudhuri–Hocquenghem (BCH) error-correction codes mechanisms in NAND flash memory.
- Reverse-engineered page-level interactions using C and microcode to map aggressor–victim relationships, achieving 50,000+ bit flips per one million write cycles.
- Demonstrated the feasibility of an end-to-end privilege escalation attack by utilizing a read-disturbance attack vector on non-volatile memory.

Wireless Information Network Laboratory (WINLAB) **Jun. 2023 - Aug. 2023**
Supervisor: Prof. Jorge Ortiz *(Rutgers University)*
Precision-Synchronized Distributed IoT Testbed for Indoor Multimodal Learning

- Built and deployed a Python/Linux experimentation platform to evaluate human interaction patterns within indoor spaces and advance multimodal learning predictions.
- Implemented and validated Precision Time Protocol (PTP) across the testbed, achieving nanosecond-level clock synchronization.
- Integrated data streams from 25 Raspberry Pis to securely communicate environment data to a Flask-based backend for Large Language Model (LLM) processing.

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Publications and Presentations

- **Jeremy Hui**, Nikhil Sampath (Jul. 2024). *Performing Rowhammer-Like Attacks on OpenSSD NAND Flash*. Presentation at Princeton University, USA (Presentation)
- Katrina Celario, **Jeremy Hui**, Julia Rodriguez, Laura Liu, Jose Rubio, Michael Cai (Aug. 2023). *Robotic IoT SmartSpace Testbed*. Poster and Presentation at WINLAB, USA (Presentation) (Poster)

Honors and Awards

- Rutgers Engineering Honors Academy, 2022-2026 (80 spots chosen from 1250+ students for academic ability, intellectual curiosity, and leadership in engineering)
- Tau Beta Pi - NJ Beta Chapter, 2024 (Top 8th of junior year class)

Teaching and Mentorship

- **Introduction to Computer Science (CS 111)** - Lead Teaching Assistant and Assignment Developer (Summer 2023, Spring 2024, Summer 2024)
  - Wrote 5 Java assignments and 2 auto-graders for 1000+ students promoting humanitarian applications of CS. Supported over 600 students through holding office hours, and coordinated logistics for 30+ TAs.
- **Linear Systems and Signals (ECE 345)** - Learning Assistant (Fall 2025)
  - Taught 50+ students through weekly active learning recitations. Created recitation problems, responded to student questions on course forums, and provided regular feedback to curricular design.
- **Intro to Data Driven Design for Engineering Applications 2 (ENG 127/102)** - Learning Assistant (Fall 2023, Spring 2024, Spring 2025)
  - Taught 80+ students MATLAB through weekly active learning recitations. Created pilot recitation assignments for 1000+ students to promote hands-on learning.
- **Data Structures (CS 112)** - Grader (Fall 2023)
  - Graded 3 exams for 100+ students.

Languages and Skills

- **Languages:** HTML, JavaScript, Java, C/C++, MATLAB, Python, SQL (MySQL, SQLite3), RISC-V
- **Frameworks:** React, Node.js, WordPress, Express.js, Jasmine
- **Tools:** AWS EC2, gem5, Linux, GitHub, Docker, AWS S3, Vim, Vivado/SystemVerilog